

# \*1: Cross product & linear transformations

•  $\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \times \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \det \left( \begin{bmatrix} \hat{i} & v_1 & w_1 \\ \hat{j} & v_2 & w_2 \\ \hat{k} & v_3 & w_3 \end{bmatrix} \right)$ , accept weirdness:

$$\hat{i} (v_2 w_3 - v_3 w_2) + \hat{j} (v_3 w_1 - v_1 w_3) + \hat{k} (v_1 w_2 - v_2 w_1)$$

constants

• = coordinates of  $\vec{P}'$ , where

$$\rightarrow |\vec{P}'| = \text{Area}_{\square} \text{ bw } \vec{v} \text{ \& } \vec{w}$$

•  $\rightarrow \vec{P}' = \perp$  to  $\square$  bw  $\vec{v}$  &  $\vec{w}$

• duality here:

$\rightarrow$  lin tra from space  $\rightarrow$  1D ~~or~~ line...

$\rightarrow$  associated w/ unique  $\vec{v}'$  in that space

$\rightarrow$  performing  $L()$ , same as taking dot product w/  $\vec{v}'$

- $\rightarrow$  bc  $L()$  described by matrix w/ just 1 row,  $[. \ .]$ , meaning basis vectors land on a number.

$$\begin{aligned} &\rightarrow \underset{\substack{\text{transform} \\ \text{matrix}}}{[. \ .]} \underset{\substack{\text{associated} \\ \text{vector}}}{[\vec{v}]} \text{ computationally identical} \\ &\quad \text{w/ } \vec{v} \cdot (\text{transform matrix on side}) \\ &= [\vec{v}] \cdot [\vec{v}] \end{aligned}$$

- when u find lin trans to number line, you can match it to some vector, the dual vector of that transformation
- where performing lin transformation = same as taking dot product w/ that vector
- cross product gives ex of this <sup>in</sup> action

plan

- 1) define  $3D \rightarrow 1D L(\cdot)$  in terms of  $\vec{v}, \vec{w}$
- 2) find its dual vector in 3D space
- 3) show that dual vector =  $\vec{v} \times \vec{w}$

• understanding transform makes clear  
computation  $\leftrightarrow$  geometry of  $X$

• 2D version of cross product:

$$\vec{v} \times \vec{w} = \det \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} = \text{Area} \begin{matrix} \square \\ \text{by } \vec{v} \text{ \& } \vec{w} \end{matrix} \text{ spanned}$$

negative  $\Rightarrow$  orientation



• you'd think:  $\vec{v} \times \vec{v} \times \vec{w} = \det \begin{pmatrix} v_1 & v_1 & w_1 \\ v_2 & v_2 & w_2 \\ v_3 & v_3 & w_3 \end{pmatrix}$   
giving Volume parallelepiped. -

• this not the 3D cross product.

• real one is  $\vec{v} \times \vec{w} = \vec{p}$ , takes in  
2 vectors & spits out a vector

• doesn't take in 3 vectors, & output a ~~ss~~

• but the student intuition is close

$f\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \det \begin{pmatrix} x & v_1 & w_1 \\ y & v_2 & w_2 \\ z & v_3 & w_3 \end{pmatrix}$ , a function  
from 3D to 1D!

= "for any 3D input vector, consider Parallelepiped  
defined by input,  $\vec{v}$  &  $\vec{w}$ , return volume w/ sign of orientation"

- this function is linear... based on properties of the determinant
- once proven linear, can use duality
- You can describe this function as matrix multiplication..

$$\begin{bmatrix} \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \det \left( \begin{bmatrix} x & v_1 & w_1 \\ y & v_2 & w_2 \\ z & v_3 & w_3 \end{bmatrix} \right)$$

1x3 matrix

encoding transformation  
from 3D  $\rightarrow$  1D

- duality here: if 3D  $\rightarrow$  1D, turn matrix on side, take dot product

$$\begin{bmatrix} ? \\ ? \\ ? \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \det \left( \begin{bmatrix} x & v_1 & w_1 \\ y & v_2 & w_2 \\ z & v_3 & w_3 \end{bmatrix} \right)$$

- we're looking for special 3D vector,  $\hat{p}$ , st taking  $\hat{p} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$  gives same result as plugging in  $x, y, z$  as 1st col

$$\underbrace{\begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}}_{\vec{p}} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \det \left( \begin{array}{ccc} x & v_1 & w_1 \\ y & v_2 & w_2 \\ z & v_3 & w_3 \end{array} \right)$$

$\vec{v} \quad \vec{w}$

- $p_1 x + p_2 y + p_3 z = x(v_2 w_3 - v_3 w_2) + y(\dots) + z(\dots)$

so

$$\begin{aligned} p_1 &= v_2 w_3 - v_3 w_2 \\ p_2 &= v_3 w_1 - v_1 w_3 \\ p_3 &= v_1 w_2 - v_2 w_1 \end{aligned}$$

- this is same as plugging in  $\hat{i}, \hat{j}, \hat{k}$  in here, signals we should interpret those coefficients as coordinates of a vector



• what vector  $\vec{p}$  has property:

$$\rightarrow \text{take } \vec{p} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \det \begin{pmatrix} x & v_1 & w_1 \\ y & v_2 & w_2 \\ z & v_3 & w_3 \end{pmatrix}$$

• Same question, geometrically

• what 3D vector  $\vec{p}$ :

$\vec{p} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$  = signed volume of parallelepiped  
delimited by  $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ ,  $\vec{v}$  &  $\vec{w}$ ?

$$\vec{p} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = |\text{Proj } P_{\vec{p}}| \cdot |p|$$

• Vol = Area  $\square$  • component of  $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$   $\perp$  to  $\square$

same as  $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \cdot \vec{v} \perp \text{ b/w } \vec{v} \propto \vec{w}$  w/ length  
= Area  $\square$

= found a vector  $\vec{p}$  s.t

$$\vec{p} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \det \begin{bmatrix} x & y & z \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}$$

$\Rightarrow$  proves geometric interpretation  
of cross product & computation  
are related.